

DUDINKA, Russian Federation

WMO: 230740

Lat: 69.400N

Lon: 86.167E

Elev: 19

StdP: 101.10

Time Zone: 7.00 (E07)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-45.7	-43.8	-48.4	0.0	-45.7	-46.4	0.0	-43.8	14.1	-14.6	12.3	-13.6	1.7	110	0.947

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	25.0	17.4	22.1	16.1	19.7	14.7	18.3	23.2	16.7	21.1	15.3	19.0	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	20.4	14.7	10.5	19.0	13.3	9.6	17.6	52.0	23.2	47.2	21.0	43.0	19.3	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.9	9.6	DB	-47.4	27.6	3.3	2.0	-49.8	29.1	-51.7	30.3	-53.5	31.4	-55.9	32.9	(4)
(5)			WB	-47.4	19.1	3.3	1.5	-49.8	20.2	-51.7	21.1	-53.5	22.0	-55.9	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.7	-26.1	-27.0	-20.5	-13.5	-4.4	8.1	13.8	11.1	5.0	-7.1	-20.3	-24.5	(6)
(7)		DBStd	16.58	9.76	9.91	9.91	8.47	6.00	5.68	4.12	3.79	4.14	7.81	9.60	9.93	(7)
(8)		HDD10.0	7063	1119	1037	945	705	448	105	8	32	156	529	911	1068	(8)
(9)		HDD18.3	9876	1377	1271	1203	955	705	312	152	227	399	787	1161	1327	(9)
(10)		CDD10.0	248	0	0	0	0	1	47	127	65	7	0	0	0	(10)
(11)		CDD18.3	18	0	0	0	0	0	4	12	2	0	0	0	0	(11)
(12)		CDH23.3	134	0	0	0	0	1	33	89	11	0	0	0	0	(12)
(13)		CDH26.7	18	0	0	0	0	0	5	13	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.7	4.7	4.2	4.6	4.7	5.0	4.8	4.9	4.6	4.8	4.9	4.3	5.0	(14)
(15)	Precipitation	PrecAvg	511	36	33	33	32	33	33	55	58	41	57	42	40	(15)
(16)		PrecMax	645	81	75	72	69	63	82	261	97	100	97	81	74	(16)
(17)		PrecMin	373	4	7	8	2	2	3	1	17	15	4	5	0	(17)
(18)		PrecStd	78	19	17	20	18	14	22	54	22	21	19	19	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.4	-4.1	0.8	5.4	15.9	27.8	28.4	25.1	19.4	10.1	1.1	-2.8	(19)
(20)			MCWB	-3.9	-4.4	-0.7	2.4	10.2	17.2	18.7	18.8	13.7	8.4	0.4	-3.3	(20)
(21)		2%	DB	-8.1	-7.1	-2.1	2.9	6.6	23.6	26.1	21.3	14.7	5.0	-0.9	-4.8	(21)
(22)			MCWB	-8.7	-7.6	-3.0	1.0	3.7	15.4	17.9	16.5	11.2	4.2	-1.5	-5.3	(22)
(23)		5%	DB	-10.9	-10.1	-4.5	1.3	4.6	20.3	23.4	19.0	12.6	3.7	-4.2	-7.8	(23)
(24)			MCWB	-11.5	-10.7	-5.1	0.3	2.3	14.1	17.0	15.1	10.0	3.0	-4.8	-8.4	(24)
(25)		10%	DB	-13.5	-12.8	-6.8	-0.7	2.8	17.1	20.7	17.0	10.8	1.8	-6.9	-10.7	(25)
(26)			MCWB	-14.0	-13.3	-7.5	-1.8	1.4	12.1	15.3	13.8	8.7	1.0	-7.5	-11.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.8	-4.6	-0.3	2.7	10.2	18.5	20.2	19.6	15.2	8.3	0.7	-3.2	(27)
(28)			MCDWB	-3.3	-4.3	0.6	5.1	15.6	24.9	25.3	23.8	19.2	9.7	1.1	-3.0	(28)
(29)		2%	WB	-8.6	-7.6	-2.8	1.4	4.1	16.4	18.9	17.1	12.1	4.3	-1.4	-5.3	(29)
(30)			MCDWB	-8.2	-7.0	-2.1	2.7	6.2	22.6	24.1	20.2	13.9	4.8	-0.8	-4.8	(30)
(31)		5%	WB	-11.4	-10.6	-5.2	0.4	2.6	14.4	17.5	15.5	10.2	3.1	-4.7	-8.4	(31)
(32)			MCDWB	-10.9	-10.0	-4.6	1.4	4.2	20.1	22.4	18.4	12.3	3.8	-4.1	-7.9	(32)
(33)		10%	WB	-14.0	-13.3	-7.4	-1.7	1.5	12.2	16.1	14.1	8.8	1.2	-7.5	-11.3	(33)
(34)			MCDWB	-13.5	-12.8	-6.8	-0.7	2.7	17.0	20.2	16.7	10.5	1.7	-6.9	-10.7	(34)
(35)	Mean Daily Temperature Range	MDBR	7.0	7.3	8.9	9.0	6.1	6.7	7.3	6.5	4.9	5.2	7.1	6.7	(35)	
(36)		5% DB	MCDBR	8.4	9.6	8.1	6.1	6.5	12.1	10.7	9.9	8.0	3.5	8.2	8.6	(36)
(37)			MCWBR	8.0	9.5	7.9	5.3	4.5	6.0	5.0	5.5	5.1	3.2	8.1	8.4	(37)
(38)		5% WB	MCDWB	8.1	9.8	8.2	6.2	6.2	10.9	9.5	8.5	7.0	3.5	8.2	8.6	(38)
(39)			MCWBR	8.1	9.6	8.0	5.6	4.7	6.2	5.0	5.2	4.9	3.3	8.1	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.216	0.249	0.305	0.328	0.353	0.370	0.400	0.393	0.324	0.250	0.201	N/A	(40)	
(41)		taud	1.856	1.959	1.997	2.040	2.097	2.344	2.290	2.258	2.401	2.084	1.883	N/A	(41)	
(42)		Ebn at Noon	16	462	674	791	818	822	775	725	693	462	14	0	(42)	
(43)		Edh at Noon	5	47	90	118	127	103	106	97	63	39	3	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.01	0.35	1.91	4.10	5.70	5.35	5.01	3.14	1.58	0.49	0.06	0.00	(44)	
(45)		RadStd	0.00	0.05	0.19	0.33	0.42	0.46	0.70	0.36	0.18	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page